



Complex Engineering Problem (CEP)

Electronic Devices and Circuits (EE-120)

Spring 2026

BS Electrical Engineering

Section A

Group Members

Student Name	Registration Number
Laiba Talat	02-3-1-038-2025
Fatima Zahoor	02-3-1-022-2025
Hooriya Zahid	02-3-1-034-2025
Malaika Tariq	02-3-1-040-2025

Instructors:

Dr. Muhammad Riaz

Dr. Muhammad Aqil

Lab Engineers:

Engr. Nouman Aftab

Engr. Muhammad Saad

Submission Date: May 12, 2026

Contents

1.	Objectives	2
2.	Abstract	2
3.	Theory	2
3.1	DC power supply	2
3.2	Transformer	2
3.3	Rectification Circuit	2
3.4	Filter Circuit	3
3.5	Voltage Regulator (Zener Diode)	3
4.	Components	3
5.	Circuit	3
5.1	Circuit in LtSpice	3
5.2	Simulations in LTSpice	4
6.	Experimental Procedure	5
7.	Experimental Results	6
8.	Calculations	7
9.	Hardware Implementation	8
10.	Load Regulation Test	9
11.	Discussion	10
12.	Safety Precautions	11
13.	Epilogue	11

1. Objectives

- To design and construct a DC power supply.
- To understand the working of rectification and filtering circuits.
- To study the application of a Zener diode as a voltage regulator.

2. Abstract

This project presents the design and implementation of a regulated DC power supply capable of providing an output of 2.5 V DC at 0.5 mA. The system utilizes a step-down transformer to reduce the input voltage from 230 Vrms AC to 12 Vrms AC. The reduced AC voltage is then converted into DC using a rectification circuit, followed by a filtering stage to minimize ripple. A Zener diode is employed as a voltage regulator to maintain a stable and constant output voltage. The overall performance of the power supply is evaluated through both simulation and practical implementation, demonstrating its effectiveness in delivering a reliable low-voltage DC output.

3. Theory

3.1 DC power supply

A DC power supply is an electronic device that converts alternating current (AC) into direct current (DC) to provide a stable voltage for electronic circuits. DC power is needed because most electronic devices, such as microcontrollers, sensors, and low-power circuits, operate reliably only with a constant, unidirectional voltage, unlike AC which continuously changes polarity.

3.2 Transformer

A transformer is a device that changes the level of AC voltage using the principle of electromagnetic induction, without altering the frequency. In a DC power supply, it is used to reduce the high mains voltage to a lower, safer AC voltage suitable for rectification and regulation. In this project, a step-down transformer reduces the 230 Vrms AC input to 12 Vrms AC, providing the proper voltage for the rectifier, filter, and Zener diode voltage regulator stages.

3.3 Rectification Circuit

The rectification circuit is responsible for converting the AC voltage into a pulsating DC. This is achieved using diodes, which allow current to flow only in one direction. A full-wave bridge rectifier, which utilizes four diodes, is typically used in such circuits to maximize the efficiency of converting the AC voltage to DC.

3.4 Filter Circuit

The filter circuit is designed to remove the ripple from the rectified DC output, which is inherently fluctuating due to the nature of the rectification process. This is typically achieved by using capacitors, which smooth out the pulsating DC waveform and provide a more stable and continuous output voltage suitable for powering sensitive components.

3.5 Voltage Regulator (Zener Diode)

A Zener diode is used as a voltage regulator in power supplies. When connected in reverse bias and operated in its breakdown region, the Zener diode maintains a constant output voltage. This feature allows it to stabilize the voltage output by absorbing excess voltage and preventing fluctuations, ensuring a steady supply of voltage for sensitive electronic circuits.

4. Components

The following components were used to achieve the objective:

- Step-Down Transformer: 12 V_{rms}
- Capacitor: 100 μ F
- Diodes: Four silicon diodes ($V_f \approx 0.5$ V each)
- Zener Diode: 2.5 V
- Resistors: 5 k Ω resistor and 1 k Ω resistor

5. Circuit

5.1 Circuit in LtSpice

The circuit is first designed and tested using simulation software.

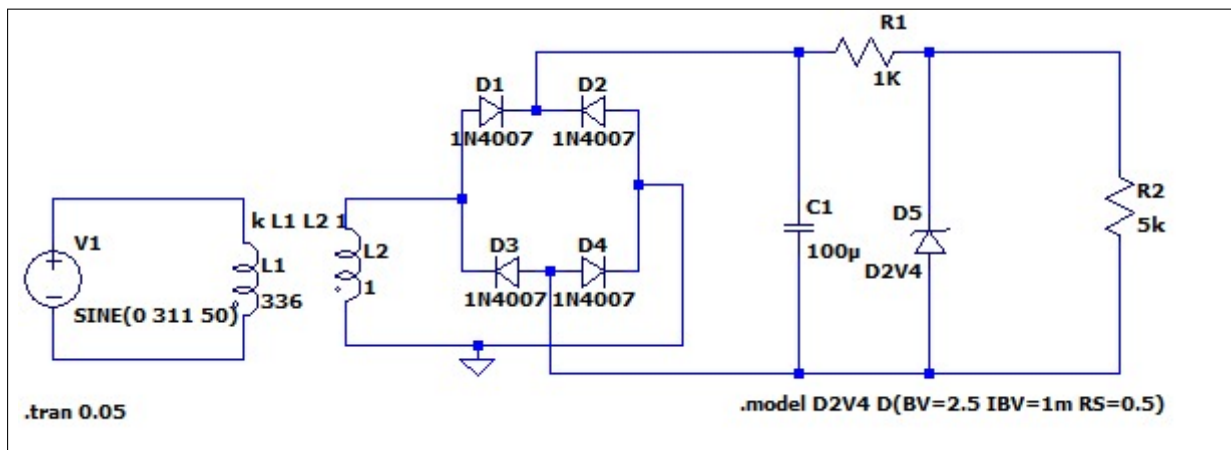


Figure 1: Regulated DC Power Supply Circuit

5.2 Simulations in LTSpice

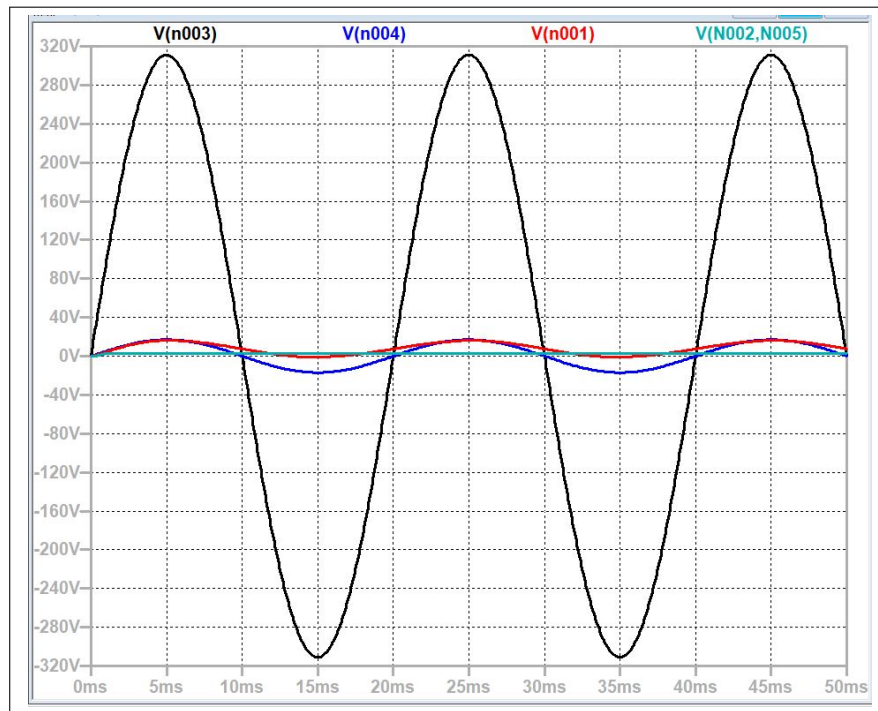


Figure 2: Overall waveform (including input of 220 V_{rms})

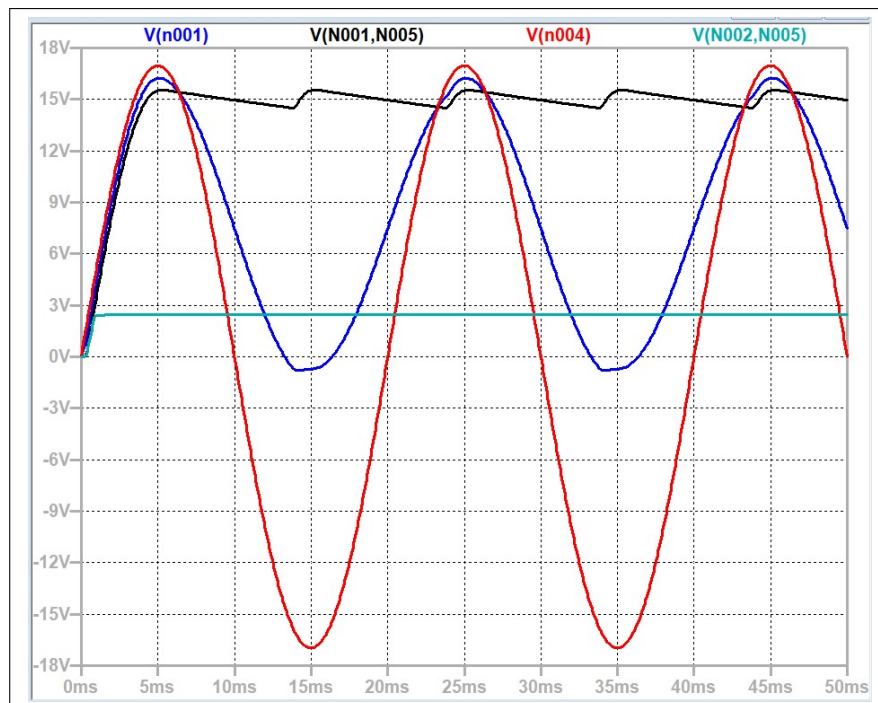


Figure 3: Zoomed in waveform (including stepped-down input of 12 V_{rms})

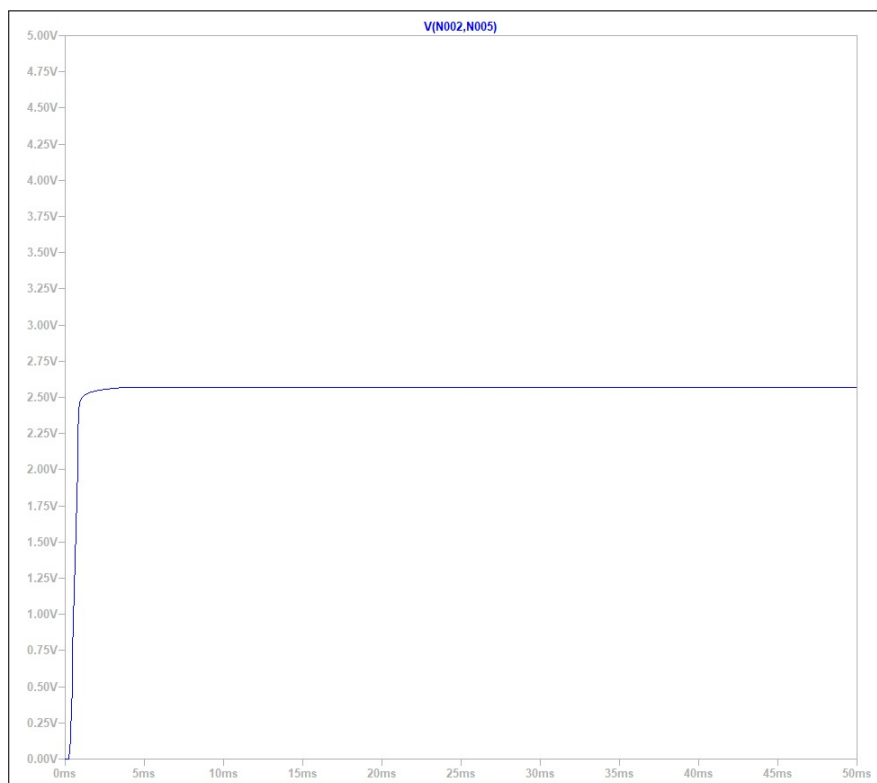


Figure 4: Output regulated at 2.5 V

6. Experimental Procedure

1. **Circuit Assembly:** Assemble the components according to the circuit diagram using the following stage-by-stage approach:
 - (a) **Transformer:** Connect the secondary windings of the transformer to the breadboard.
 - (b) **Rectification:** Connect the transformer output to the input terminals of the Full-Wave Bridge Rectifier.
 - (c) **Filtering:** Connect the filter capacitor in parallel across the rectifier output, ensuring correct polarity.
 - (d) **Regulation:** Connect a series current-limiting resistor (R_s) from the positive filter rail to the cathode of the Zener diode. Ground the anode.
 - (e) **Load:** Connect the load resistor (R_L) in parallel with the Zener diode.
2. **Measurements using Digital Multimeter (DMM)** Apply power to the circuit and measure the following parameters:
 - **Input Voltage (V_{in}):** Measure the AC voltage at the transformer secondary.
 - **Unregulated DC:** Measure the DC voltage across the filter capacitor.
 - **Regulated Output (V_{out}):** Measure the DC voltage across the load resistor to verify it matches the Zener voltage rating.
3. **Oscilloscope Observation** Connect the oscilloscope probes to visualize the signal transformation:

- Observe the **Pulsating DC** at the output of the rectifier (before the capacitor).
- Observe the **Filtered DC** and measure the ripple voltage across the capacitor.
- Observe the **Regulated DC** across the load, noting the stability of the output line.

7. Experimental Results

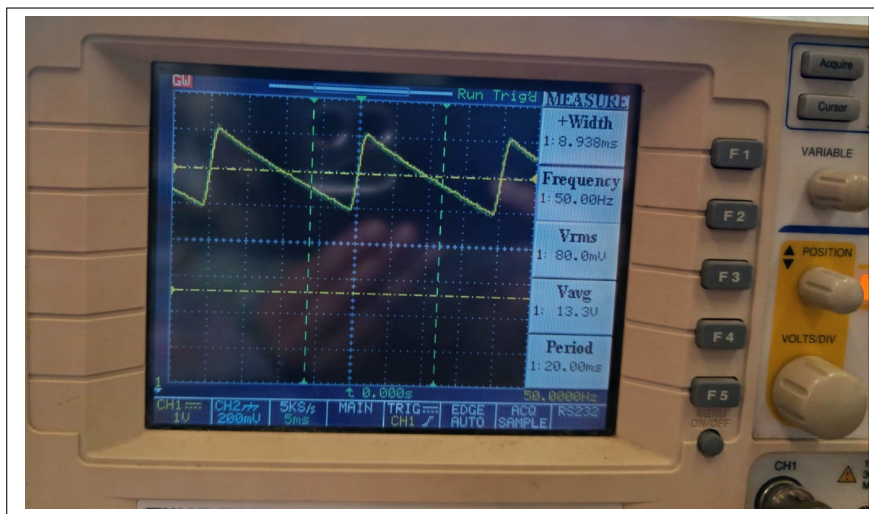


Figure 5: Capacitor filtering with 2% ripple

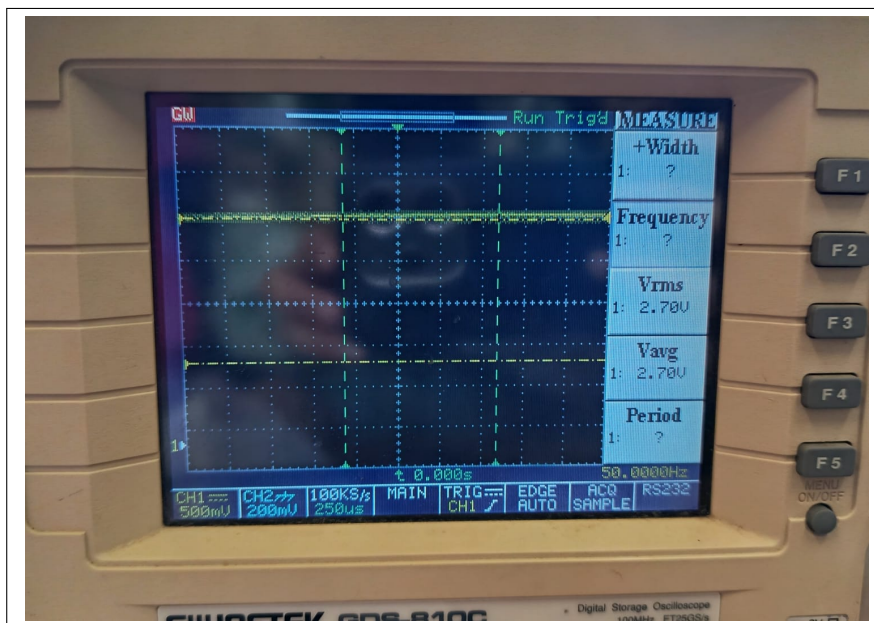


Figure 6: Final output showing regulated DC at approx 2.5V

8. Calculations

Current and Capacitance Calculation:

Given:

$$V_{\text{out}} = 2.5 \text{ V}$$

$$R_{\text{load}} = 5 \text{ k}\Omega$$

$$\text{Ripple Percentage} = 2\%$$

Ohm's Law:

$$I = \frac{V}{R}$$

$$I = \frac{2.5}{5000}$$

$$I = 0.5 \text{ mA}$$

$$f = 50 \text{ Hz}$$

For a full-wave rectifier:

$$f_r = 2f = 100 \text{ Hz}$$

First, calculate ripple voltage:

$$V_{\text{ripple}} = \left(\frac{2}{100} \right) \times 2.5$$

$$V_{\text{ripple}} = 0.05 \text{ V}$$

Using ripple voltage formula:

$$V_{\text{ripple}} = \frac{I_{\text{load}}}{f_r C}$$

Rearranging for capacitance:

$$C = \frac{I_{\text{load}}}{f_r V_{\text{ripple}}}$$

Substituting values:

$$C = \frac{0.5 \times 10^{-3}}{100 \times 0.05}$$

$$C = 1 \times 10^{-4} \text{ F}$$

$$C = 100 \mu\text{F}$$

9. Hardware Implementation

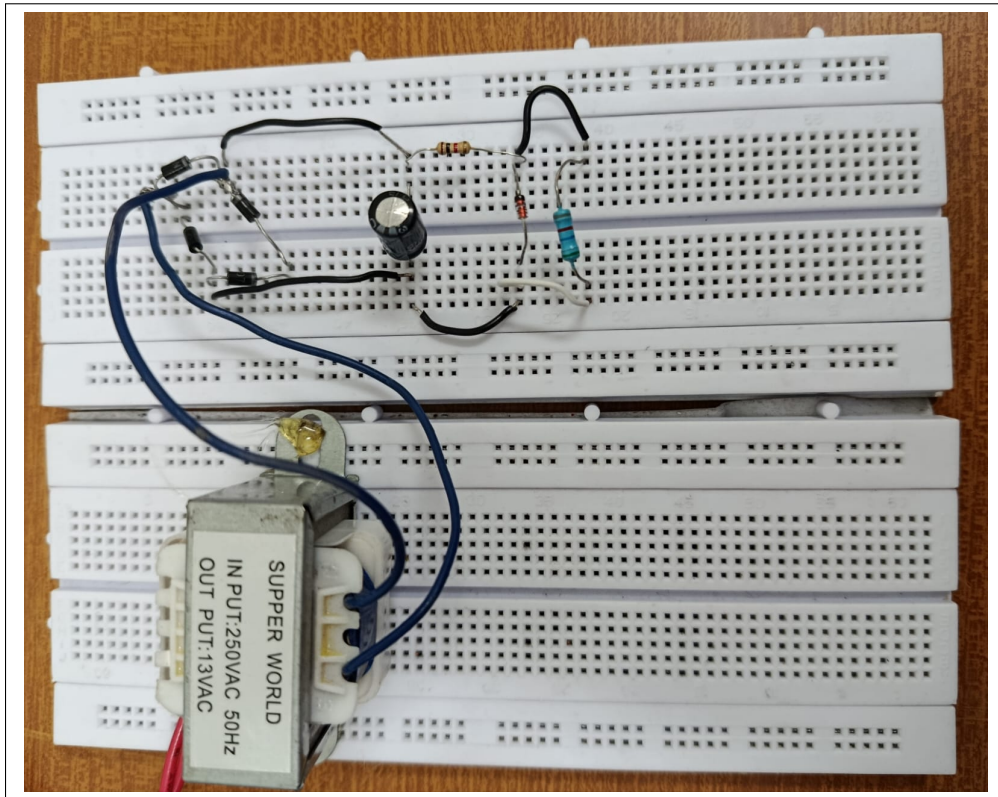


Figure 7: Regulated DC Power Supply Circuitry

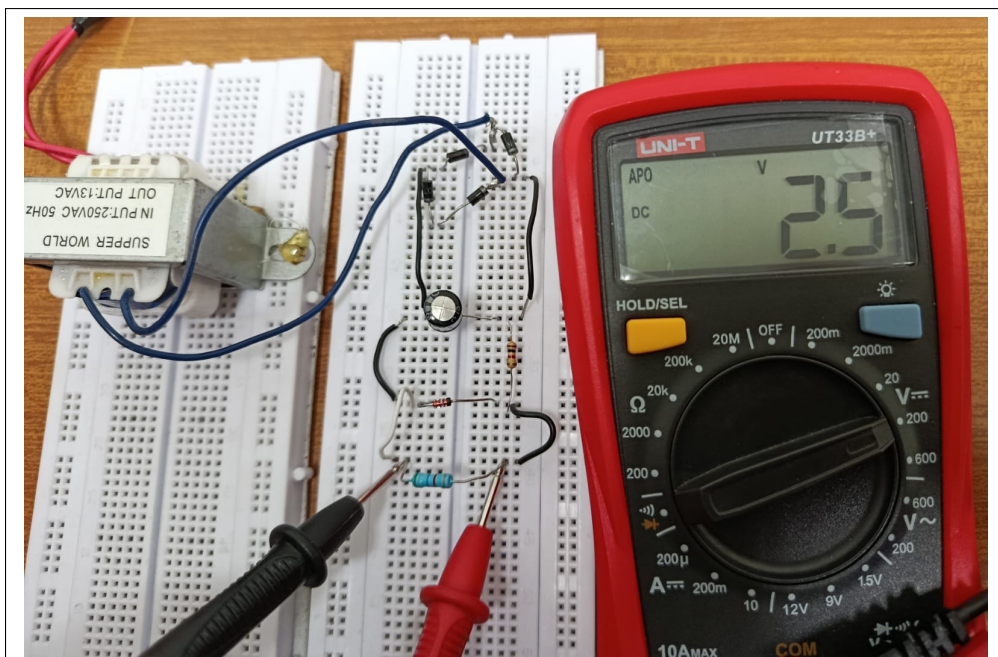


Figure 8: Regulated DC Voltage across 5k Ω load

10. Load Regulation Test

Vary the load resistor (R_L) and observe if the output voltage V_{out} remains constant, demonstrating the regulation effect of the Zener diode.

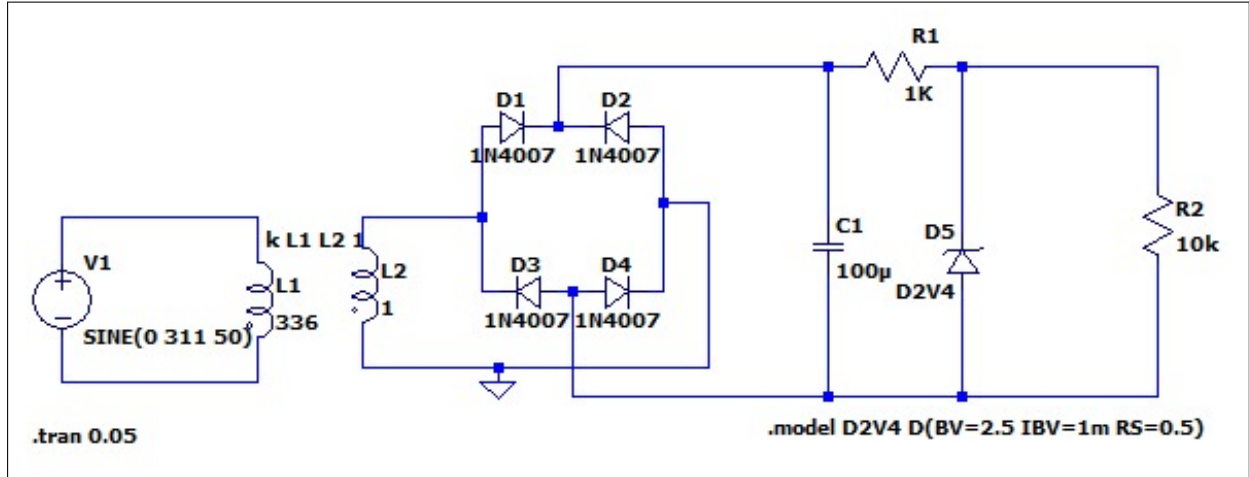


Figure 9: Zoomed in waveform (including stepped-down input of 12 V_{rms} using 10 kΩ load)

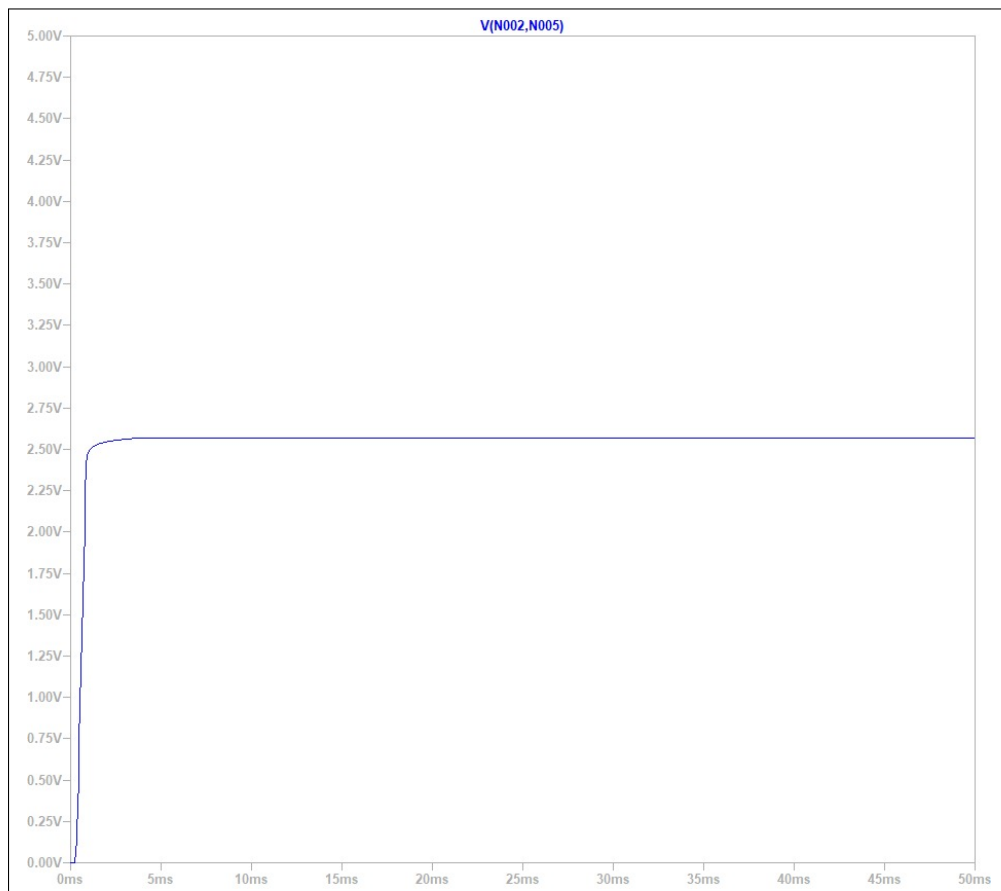


Figure 10: Output regulated at 2.5 V using 10 kΩ load

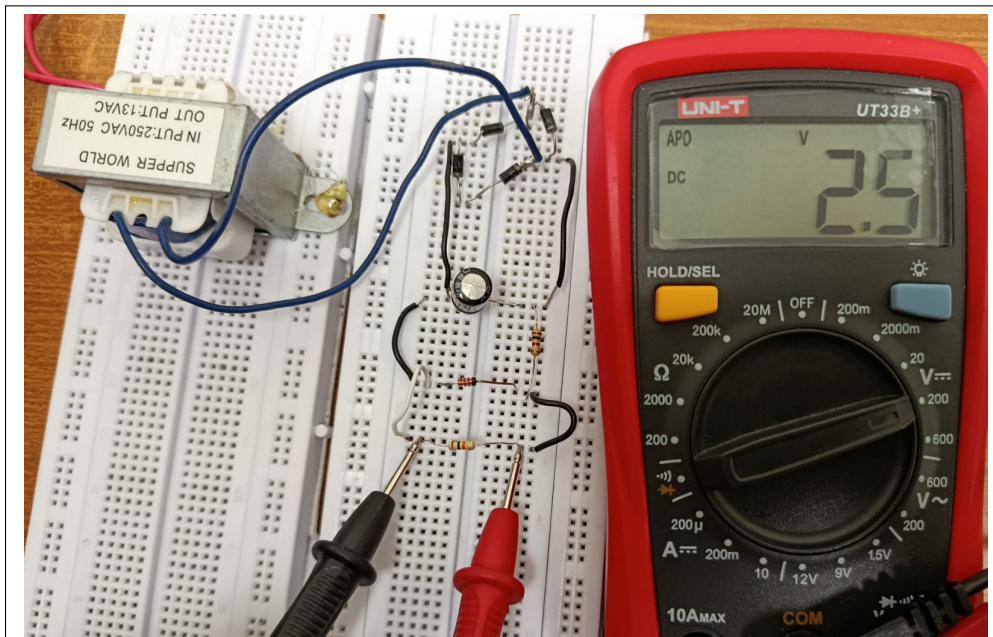


Figure 11: Load regulation at 10k Ω load

11. Discussion

The objective of this project was to design and implement a regulated DC power supply capable of producing a stable 2.5 V output. The complete system consisting of a transformer, bridge rectifier, filter capacitor, and Zener diode regulator was successfully analyzed through simulation and verified experimentally.

The transformer effectively stepped down the 230 V AC mains supply to 12 V AC, which was then converted into pulsating DC by the full-wave bridge rectifier. The filter capacitor significantly reduced the ripple content of the rectified signal, resulting in a smoother DC waveform. Measurements and oscilloscope observations confirmed that the ripple voltage remained within the desired 2% limit.

The Zener diode regulator maintained the output voltage close to the desired value of 2.5 V under different load conditions. Both LTSpice simulations and practical hardware implementation produced output voltages in the range of 2.5 V to 2.7 V, demonstrating good agreement between theoretical calculations and experimental results.

The load regulation test further showed that the output voltage remained relatively constant when the load resistance was varied, confirming the effectiveness of the voltage regulation stage. Minor variations in output voltage were mainly due to component tolerances, diode voltage drops, and practical limitations of the hardware setup.

Overall, the project successfully demonstrated the principles of AC to DC conversion, rectification, filtering, and voltage regulation. It also provided valuable practical experience in circuit design, simulation, hardware implementation, and measurement techniques used in basic electronic power supply systems.

12. Safety Precautions

1. Ensure the power supply is switched OFF before making or changing circuit connections.
2. Verify correct polarity of components such as electrolytic capacitors, diodes, and voltage regulators before powering the circuit.
3. Do not exceed the rated input voltage, current, or power limits of the regulated power supply.
4. Use properly insulated wires and avoid touching exposed conductors while the circuit is energized.
5. Check all connections carefully to prevent short circuits and accidental damage to components.
6. Discharge filter capacitors safely before handling the circuit after power is turned OFF.
7. Keep the working area dry and free from conductive materials to avoid electric shock hazards.

13. Epilogue

In conclusion, the regulated DC power supply was successfully designed, simulated, and implemented. The circuit effectively converted AC voltage into a stable 2.5 V DC output using rectification, filtering, and Zener diode regulation techniques. Both simulation and hardware results confirmed proper operation with reduced ripple and satisfactory load regulation. This project provided practical understanding of power supply design and the application of fundamental electronic components in voltage.